

Safety and Efficacy of the SNAP 12-hour Acetylcysteine Regimen for the Treatment of Paracetamol Overdose.

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Acetylcysteine (NAC) is effective at preventing liver injury after paracetamol overdose. The Scottish and Newcastle Anti-emetic Pre-treatment for Paracetamol Poisoning (SNAP) Study demonstrated that a 12 h NAC regimen was associated with fewer adverse drug reactions compared with the standard 21 h regimen. Here, we describe the clinical effectiveness of the SNAP NAC regimen. The SNAP regimen, consisting of intravenous NAC 100 mg/kg over 2 h then 200 mg/kg over 10 h, was introduced to treat all paracetamol overdose patients at the Royal Infirmary of Edinburgh, the Royal Victoria Infirmary, Newcastle and St Thomas' Hospital, London. Patient data were prospectively and systematically collected before and after the change in treatment (total patients N = 3340, 21 h N = 1488, SNAP N = 1852). Health record linkage was used to determine patient outcome after hospital discharge. There was no difference in liver injury or liver synthetic dysfunction between regimens. Hepatotoxicity (peak ALT > 1000 U/L) occurred in 64 (4.3%) and 67 (3.6%) patients, respectively, in the 21 h and SNAP groups (absolute difference - 0.7%, 95% CI - 2.1 to 0.6). Multivariable logistic regression did not identify treatment regimen as an outcome-associated factor. No patients were readmitted to hospital with, or died from, liver failure within 30 days of discharge. Anti-histamine treatment (for NAC anaphylactoid drug reactions) was prescribed for 163 (11.0%) patients with the 21 h regimen and 37 (2.0%) patients with the SNAP regimen (absolute difference 9.0% (95% CI 7.3 to 10.7)). In clinical use the SNAP regimen has similar efficacy as standard therapy for preventing liver injury and produces fewer adverse reactions.

Delayed presentation of paracetamol overdose.

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Deliberate self-harm, including intentional self-poisoning, remains a major health concern in South Africa. Increasing number of cases place a significant burden on our Emergency Centres (EC's). Paracetamol remains the most frequent drug ingested in intentional self-poisoning. As an antidote, N-acetylcysteine (NAC) is very effective in the management of acute cases of paracetamol overdose. However, it shows less efficacy in cases of delayed presentation, where both liver transplant and mortality rates are significantly higher. We present a case of delayed presentation paracetamol overdose. The patient presented in fulminant hepatic failure with encephalopathy, but made a full recovery after being treated with NAC. This case report highlighted NAC's potential effectiveness in delayed presentations of paracetamol overdose, irrespective of associated fulminant hepatic failure. The effectiveness of NAC in delayed presentations of paracetamol overdose should therefore not be underestimated, and warrants further research.

[Clinical-toxicological case (1). Dosage of N-acetylcysteine in acute paracetamol poisoning].

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There are currently three protocols used for the administration of N-acetylcysteine in the treatment of acute paracetamol poisoning. In the USA only the oral protocol is approved, while in Europe an

intravenous protocol is used. If treatment is started within 10 h. after paracetamol ingestion, all three protocols appear to be equally effective. If treatment is started 10 to 24 h. after the ingestion, the oral protocol and the Smilkstein protocol appear to be superior to the Prescott protocol. N-acetylcysteine is effective also when started more than 15 h after the ingestion. Patients who present with liver failure after paracetamol poisoning should be treated with a prolonged course of N-acetylcysteine.

Effect of acetylcysteine on prothrombin index in paracetamol poisoning without hepatocellular injury.

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Acetylcysteine treatment reduces liver damage after paracetamol overdose, but can affect the prothrombin index, which is used to assess the progress of overdose patients. We aimed to assess retrospectively the effect of intravenous acetylcysteine on the prothrombin index in patients with paracetamol poisoning without signs of hepatocellular injury. Prothrombin index had been recorded before, and serially during, acetylcysteine treatment in 87 patients. After initiation of treatment, prothrombin index decreased (mean 0.33, 95% CI 0.29-0.38) in all patients, and was strongly associated with the start of acetylcysteine infusion. In patients with uncomplicated paracetamol poisoning, a fall in this index might be misinterpreted as a sign of liver failure, leading to prolonged treatment time.

Growth Hormone Accelerates Recovery From Acetaminophen-Induced Murine Liver Injury.

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Acetaminophen (APAP) overdose is the leading cause of acute liver failure, with one available treatment, N-acetyl cysteine (NAC). Yet, NAC effectiveness diminishes about ten hours after APAP overdose, urging for therapeutic alternatives. This study addresses this need by deciphering a mechanism of sexual dimorphism in APAP-induced liver injury, and leveraging it to accelerate liver recovery via growth hormone (GH) treatment. GH secretory patterns, pulsatile in males and near-continuous in females, determine the sex bias in many liver metabolic functions. Here, we aim to establish GH as a novel therapy to treat APAP hepatotoxicity. Our results demonstrate sex-dependent APAP toxicity, with females showing reduced liver cell death and faster recovery than males. Single-cell RNA sequencing analyses reveal that female hepatocytes have significantly greater levels of GH receptor expression and GH pathway activation compared to males. In harnessing this female-specific advantage, we demonstrate that a single injection of recombinant human GH protein accelerates liver recovery, promotes survival in males following sub-lethal dose of APAP, and is superior to standard-of-care NAC. Alternatively, slow-release delivery of human GH via the safe nonintegrative lipid nanoparticle-encapsulated nucleoside-modified mRNA (mRNA-LNP), a technology validated by widely used COVID-19 vaccines, rescues males from APAP-induced death that otherwise occurred in control mRNA-LNP-treated mice. Our study demonstrates a sexually dimorphic liver repair advantage in females following APAP overdose, leveraged by establishing GH as an alternative treatment, delivered either as recombinant protein or mRNA-LNP, to potentially prevent liver failure and liver transplant in APAP-overdosed patients.

An update of N-acetylcysteine treatment for acute acetaminophen toxicity in children.

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Acetaminophen poisoning accounts for a disproportionate percentage of all toxic ingestions, and can be life-threatening. This article reviews the mechanism and presentation of acetaminophen toxicity, as well as its treatment, including current thinking and treatment recommendations. N-acetylcysteine acts to detoxify acetaminophen in several ways, but primarily by increasing the synthesis and availability of glutathione, which binds and inactivates the highly reactive and hepatotoxic acetaminophen metabolite N-acetyl-p-benzoquinoneimine. The US Food and Drug Administration has approved an intravenous formulation of N-acetylcysteine, thus allowing the treatment time to be decreased from the 72 hr most commonly used for the oral regimen, to only 20 hr. This comes after many years of accepted intravenous N-acetylcysteine use in Europe and Canada, and much controversy as to the superiority of both treatments. This review summarizes this controversy, and offers a framework to develop a safe treatment plan that has the optimal outcome for the patient, as well as reflecting knowledge of the potential caveats at work. It describes side effects of N-acetylcysteine treatment, as well as relative indications to choose one route of treatment over the other. Acetaminophen can lead to irreversible liver damage and even death in acute overdose. Outcome is related to the swiftness in which the antidote (N-acetylcysteine) is provided. In the United States, there are now available both the oral and intravenous forms of N-acetylcysteine, and pros and cons exist for each. With brisk and adequate treatment using either route, recovery can be complete, and liver function can be restored.

N-Acetylcysteine (NAC) for the Prevention of Liver Failure in Heat Injury-Mediated Ischemic Hepatitis.

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Exertional Heat Illness with associated ischemic hepatitis (IH) is a common occurrence among military trainees; however, few specific therapies exist if unresponsive to appropriate supportive measures. A 27-year-old basic combat trainee presented with altered mental status, renal insufficiency, rhabdomyolysis, and a core temp of 107.9 °F after collapsing during a run, leading to the diagnosis of heat stroke. While the patient's azotemia and creatinine kinase levels rapidly improved with aggressive intravenous hydration, transaminases continued to increase to nearly 155 times the upper limit of normal. Further laboratory evaluation revealed coagulopathy and thrombocytopenia suggestive of acute liver failure (ALF). On hospital day three, the patient was started on N-acetylcysteine (NAC). Evaluation for infectious and autoimmune etiologies of ALF was unremarkable; thus, the patient's symptomatology was attributed to IH resulting from heat stroke. Liver function normalized on NAC. Heat Injury is common among US Army recruits and results in thousands of hospitalizations in recent years. IH is characterized by diffuse hepatocyte necrosis following an episode of hemodynamic instability, and is an established sequela of Heat Injury. The mortality of IH among critically ill patients has been estimated to be as high as 60%, with those demonstrating coagulopathy especially at risk. NAC is shown to improve the transplant-free survival rate in non-acetaminophen related ALF, consistent with its proposed mechanisms of improving hepatic blood flow and conjugating toxic metabolites. NAC therapy should be considered early in the course of heat injury-mediated IH to reduce reperfusion injury, improving transplant free outcomes.

Bilaterality of varicocele: The overlooked culprit in male infertility. Case series study.

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Varicocele is the most common correctable cause of male infertility that always has been a debatable subject as regards how it affects fertility and the best way to treat it. Proper assessment of the disease bilaterality is crucial not to miss one side and not to jeopardize treatment outcome. This study aimed to objectively assess varicocele bilaterality in infertile men aiming to improve treatment outcome in this cohort of patients. This prospective study was conducted between January 2019 and January 2022 including infertile males with varicoceles. Assessment of missed concomitant contralateral varicocele done pre-operatively by Color Doppler Ultrasound and intraoperatively by intraoperative Doppler device and measurement of maximal vein diameter of contralateral side. A total of 329 cases completed the study. A hundred cases (30.4%) were initially referred as unilateral varicoceles and 229 (69.6%) as bilateral varicoceles. After reassessment of the study population, bilaterality of varicocele was found to be as high as 98.5% (324/329). Repeat CDUS strongly correlated with the intraoperative measured varicocele diameter ($r = 0.9$, $p < 0.001$). Moreover, sperm parameters showed significant improvement 3 and 6 months post varicocelectomy. Normal pregnancy after 1 year of surgery occurred in 118 cases (35.9%). Varicocele bilaterality in infertile men is underreported. Thorough assessment by expert radiologists and andrologists is of paramount importance not to miss significant pathology or hazard treatment outcome.

[Varicocele and male infertility].

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Varicocele is the most common correctable cause of male infertility. It was the subject of recent Association française d'urologie (AFU) Comité d'andrologie et de médecine sexuelle (CAMS) recommendations. Since then, the literature has provided additional information. This review will comprehensively reassess current indications for the treatment of varicocele, and revisit contemporary issues in the light of current advances. Update of the literature search carried out as part of the CAMS recommendations for the period between 2020 and 2023. Microsurgical sub-inguinal varicocelectomy remains the surgical treatment of choice for infertile men with clinical varicocele and abnormal sperm parameters. It offers recurrence rates of less than 4%. It significantly improves both natural and in vitro fertilization live birth and pregnancy rates, as well as sperm count, total and progressive motility, morphology and DNA fragmentation rates. All in all, it modifies the MPA strategy in around one in two cases. Varicocele grade and bilaterality are predictive of improved sperm parameters and pregnancy rate. Treatment of subclinical varicocele is not recommended. Complications are rare, notably hydroceles (0.5%), unilateral testicular atrophy due to arterial damage (1/1000), hematomas, delayed healing and postoperative pain. Retrograde embolization is an alternative to surgery. Whenever possible, the urologist should present and discuss treatment options for varicocele with the MPA team and the patient, taking a personalized approach.

Varicocele is underdiagnosed in men evaluated for infertility: Examination of multi-center large-scale electronic health record data.

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Varicocele is the most common reversible cause of male infertility, affecting up to 20% of healthy men and 40% of men with primary infertility. The objective of this study was to investigate the prevalence of varicocele in men evaluated for infertility, and to determine rates of subsequent varicocele repair. Since reproductive endocrinologists are the first specialists seen for male infertility care in North America, we hypothesized that varicocele would be underdiagnosed when compared to its reported prevalence among men with infertility. TriNetX, a large, multicenter electronic health record (EHR) database was queried to establish a cohort of all men (above 18 years of age) with a diagnosis of male infertility. This cohort was used to identify those with ensuing varicocele diagnosis. Men who received varicocelectomy or venous embolization after a diagnosis of varicocele were then identified. Out of 101,309 men with a diagnosis of male infertility in the network, only 9768 (9.6%) had a diagnosis of varicocele. Mean age of men with varicocele was 34. Varicocelectomy or venous embolization was performed in 1699 (20.2%) and 69 (0.76%) of men with varicocele, respectively. In this cross-sectional EHR study, varicocele was underdiagnosed in men evaluated for infertility when compared with prior epidemiological studies.
